

PS8_Vincent

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1 OLS Estimation and Comparison with True Beta

The OLS estimates of β are very close to the true values, with minor differences in decimal places. The estimates for β obtained using the L-BFGS and Nelder-Mead algorithms are nearly identical, with only slight differences. Both optimization methods yield very similar results, indicating that either can be used to estimate β with negligible differences in the outcomes.

black	black
black	black ⁽¹⁾
X1	1.501
	(0.002)
X3	-1.001
	(0.002)
X4	-0.252
	(0.002)
X5	0.749
	(0.002)
X6	3.501
	(0.002)
X7	-2.001
	(0.002)
X8	0.499
	(0.002)
X9	1.003
	(0.002)
X10	1.247
	(0.002)
X11	2.001
	(0.002)
Num.Obs.	100000
R2	0.991
R2 Adj.	0.991
AIC	145143.6
BIC	145248.3
Log.Lik.	-72560.811
RMSE	0.50